

Unsupervised Visual Anomaly Detection in Industrial Components Using PatchCore: A Multi-Category Empirical Study

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Abstract — Automated visual inspection is a critical requirement for modern manufacturing quality control, where manual inspection is slow, inconsistent, and costly to scale. This paper presents an empirical study of an unsupervised anomaly detection pipeline built on PatchCore [1], applied across four structurally distinct categories of the MVTec Anomaly Detection (MVTec AD) benchmark [2]: metal nut, screw, cable, and transistor. The system uses a frozen, ImageNet-pretrained [3] Wide ResNet-50 [4] backbone to extract multi-scale patch embeddings, a greedy k-center coreset selection algorithm [5] to construct a compact memory bank of normal-appearance features, and k-nearest-neighbor distance scoring to detect and localize anomalies. We document a complete empirical development process across all four categories, including two methodological refinements discovered during evaluation: a top- k score aggregation strategy that stabilizes image-level classification against outlier patches, and a per-spatial-location score calibration technique that substantially improves detection of positionally-fixed defects. The final system achieves image-level AUROC scores of 0.963 (metal nut), 0.792 (screw), 0.935 (cable), and 0.947 (transistor), with pixel-level localization AUROC exceeding 0.83 in all categories. We analyze category-specific failure modes — including subtle surface scratches, thread-level defects, and relational component-swap defects — and discuss their implications for practical deployment.

Keywords — Anomaly detection, defect detection, unsupervised learning, computer vision, industrial quality control, PatchCore, MVTec AD, coreset selection, convolutional neural networks.

1 Introduction

Industrial quality control increasingly relies on computer vision to detect manufacturing defects that are costly to catch manually and expensive to miss. Supervised defect classifiers require large labeled datasets of every defect type, which is often impractical: defective samples are rare relative to normal production, and novel defect types can appear that were never labeled during training. Unsupervised anomaly detection addresses this by learning only the distribution of defect-free (“normal”) samples and flagging deviations at inference time [2].

This paper documents the design, implementation, and iterative empirical refinement of an anomaly detection system applied to four categories from the MVTec AD dataset [2]: metal nut, screw, cable, and transistor. These categories were chosen deliberately to span a range of industrial defect characteristics — from purely local surface damage to positionally-relational and whole-object defects — allowing the study to surface concrete, generalizable failure modes rather than a single point result.

2 Related Work

The MVTec AD dataset [2] established a standard benchmark for unsupervised, pixel-precise anomaly detection across fifteen object and texture categories, each

providing defect-free training images and a held-out test set containing both normal and multiple defect subtypes with pixel-level ground truth masks.

Reconstruction-based approaches, such as autoencoders and generative adversarial networks, detect anomalies via reconstruction error, but often struggle to localize subtle defects precisely. Feature-embedding approaches instead compare deep features of a test image against a reference distribution of normal features. PatchCore [1] represents the state of the art in this family: it extracts mid-level patch features from a frozen, ImageNet-pretrained [3] convolutional backbone (building on deep residual architectures [7]), builds a compact memory bank of representative normal patch embeddings via greedy coreset subsampling [5], and scores new patches by their nearest-neighbor distance to this memory bank. We adopt this architecture directly, using Wide ResNet-50 [4] as the backbone.

3 Methodology

3.1 System Architecture

For each category, the pipeline performs four stages: (i) feature extraction using intermediate layers of a frozen Wide ResNet-50 [4], concatenated across two spatial scales; (ii) memory bank construction via greedy k -center coreset selection [5] over all patch embeddings

from the defect-free training split; (iii) inference, in which a test image’s patch embeddings are compared against the memory bank via k -nearest-neighbor distance to produce a full-resolution anomaly map; and (iv) an aggregation step converting the anomaly map into a single image-level anomaly score.

3.2 Score Aggregation and Position Calibration

Two methodological refinements were developed empirically across the course of this study. First, aggregating the image-level score as the mean of the top- k highest-scoring patches, rather than a single maximum, substantially reduces sensitivity to isolated noisy patches (e.g. lighting artifacts) while retaining sensitivity to small defect regions. Formally, given a patch anomaly map $M \in R^{h \times w}$, the image score is

$$s = \frac{1}{k} \sum_{i=1}^k M_{(i)} \quad (1)$$

where $M_{(i)}$ denotes the i -th largest value in M .

Second, for categories photographed in a fixed pose (cable, transistor, and — retrospectively — screw), we introduce per-spatial-location score calibration: the mean $\mu_{(x,y)}$ and standard deviation $\sigma_{(x,y)}$ of the anomaly map are computed independently at each spatial position across the normal training set, and test-time scores are normalized as

$$\hat{M}_{(x,y)} = \frac{M_{(x,y)} - \mu_{(x,y)}}{\sigma_{(x,y)}}. \quad (2)$$

This addresses a specific failure mode in which certain spatial regions (e.g. component edges or a screw’s threaded tip) exhibit elevated anomaly scores in every image due to geometric or viewpoint effects, independent of true defect status.

3.3 Decision Threshold Selection

Rather than defining a classification threshold as the maximum anomaly score observed among normal test images — a heuristic sensitive to a single outlier sample — we select the operating threshold using Youden’s J statistic [6], which identifies the point on the receiver operating characteristic (ROC) curve maximizing $J = \text{TPR} - \text{FPR}$.

4 Experimental Setup

All experiments were conducted on CPU hardware. The backbone used intermediate feature maps from `layer2` and `layer3` of Wide ResNet-50, pretrained on ImageNet [3]. Images were resized to 256×256 and center-cropped to 224×224 , normalized using standard ImageNet statistics. Coreset compression targeted approximately 5,000 memory bank patches per category with

a 0.25 selection ratio applied to a capped 20,000-patch candidate pool. No data augmentation was applied during training, consistent with the requirement that the memory bank represent only the true distribution of normal appearance.

5 Results

5.1 Cross-Category Overview

Figure 1 summarizes image- and pixel-level AUROC across all four categories. Metal nut, cable, and transistor achieve image-level AUROC above 0.93, while screw — discussed in Section 5.3 — is the most challenging category at 0.792.

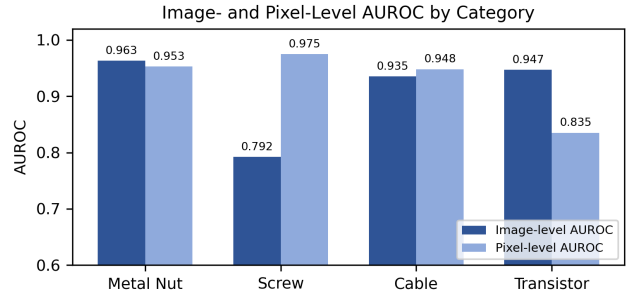


Figure 1: Image- and pixel-level AUROC across all four categories.

5.2 Metal Nut

The metal nut category (train: 220 defect-free images; test defect types: bent, color, flip, scratch) was the first implemented. Four iterations were required to reach a stable configuration (Fig. 2): an initial run with a mis-calibrated coreset ratio (0.948 AUROC, coincidentally strong despite the bug), two intermediate runs isolating the effect of memory bank size using max-based scoring (0.855, 0.837), and a final run introducing top-5 mean aggregation (0.963).

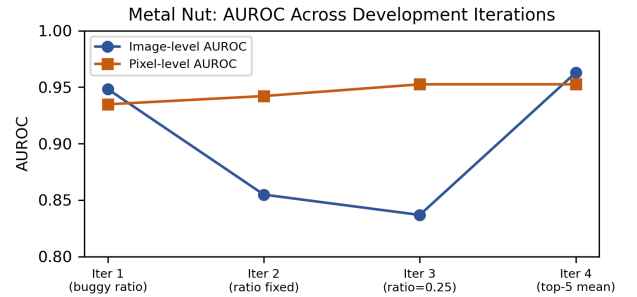


Figure 2: Metal nut AUROC across four development iterations.

The final configuration achieved image-level AUROC of 0.963 and pixel-level AUROC of 0.953. Per-defect

accuracy (Fig. 3) was 100% for bent and flip, 95.5% for color, and 73.9% for scratch — the model’s principal weakness, attributable to the low visual contrast of fine surface scratches relative to the mid-level features used for detection.

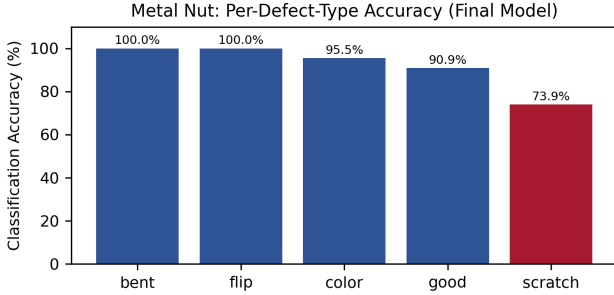


Figure 3: Metal nut per-defect-type classification accuracy, final model.

5.3 Screw

The screw category (train: 320 images; defect types: manipulated_front, scratch_head, scratch_neck, thread_side, thread_top) proved substantially more difficult. An initial run using the metal nut configuration achieved only 0.745 image-level AUROC, with thread-level defects essentially indistinguishable from normal samples in mean score. This was attributed to two compounding factors: (i) the screw’s free rotation during imaging causes a physical defect to appear at a different image location in every sample, undermining fixed-position assumptions, and (ii) thread defects are small enough that top- k mean aggregation dilutes their signal against normal patches.

Introducing per-spatial-location calibration combined with a wider top- k value ($k = 20$, selected via a monotonic sweep from $k = 1$ to $k = 1000$ that peaked near $k = 20$) improved image-level AUROC to 0.792, at the cost of an elevated false-positive rate (34.1%) on normal samples — a real precision/recall tradeoff rather than a resolved weakness. Figure 4 shows the resulting mean anomaly score separation by defect type.

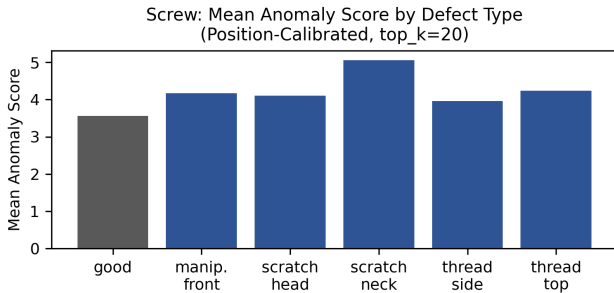


Figure 4: Screw mean anomaly score by defect type after calibration.

5.4 Cable

The cable category (train: 224 images; defect types: bent_wire, cable_swap, combined, cut_inner_insulation, cut_outer_insulation, missing_cable, missing_wire, poke_insulation) used position calibration from the first iteration, informed directly by the screw findings. This produced a strong initial result of 0.935 image-level AUROC and 0.948 pixel-level AUROC without further tuning. Per-type accuracy (Fig. 5) exceeded 90% for six of eight defect types.

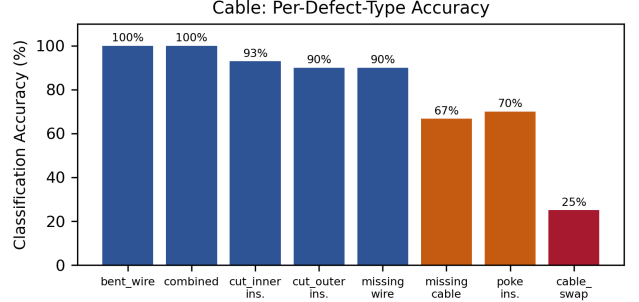


Figure 5: Cable per-defect-type classification accuracy.

The exception was cable_swap (25% accuracy), in which two individually normal-appearing wires are exchanged in position — a relational defect that patch-similarity scoring is structurally poorly suited to detect, since neither swapped patch is locally anomalous in isolation. A k -sweep ($k = 2$ to $k = 5$) confirmed this was not a tunable weakness: accuracy on cable_swap remained essentially unchanged across all values tested.

5.5 Transistor

The transistor category (train: 213 images; defect types: bent_lead, cut_lead, damaged_case, misplaced) also used position calibration from the first iteration and required no further tuning, achieving 0.947 image-level AUROC. Notably, *misplaced* — a whole-object positional defect analogous in structure to cable_swap — was the best-detected defect type (Fig. 6), confirming a hypothesis formed directly from the cable results: whole-object positional shifts produce a broad anomaly signal well-suited to top- k mean aggregation, in contrast to cable_swap’s localized two-region exchange, which does not.

Pixel-level AUROC (0.835) was lower than in other categories, plausibly because the thin, elongated geometry of lead defects is harder to localize precisely than broader surface anomalies, even when image-level detection is confident.

6 Discussion

Across all four categories, three recurring patterns emerge. First, image-level score aggregation strategy

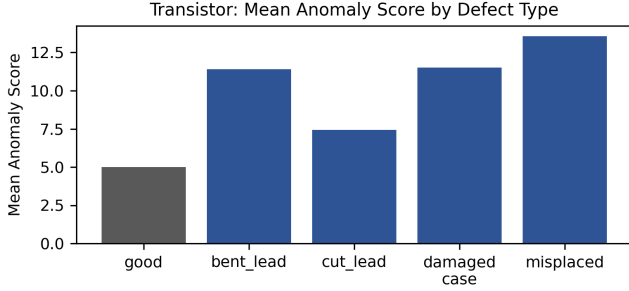


Figure 6: Transistor mean anomaly score by defect type.

materially affects classification performance independent of underlying feature quality; a single maximum score is consistently outperformed by a top- k mean, and the optimal k depends on the physical scale of the category’s defects (small for scratches, large for diffuse thread damage). Second, per-spatial-location calibration is essential for categories with a fixed imaging pose and offers limited benefit for categories with free rotation, since the underlying assumption — that a given pixel position corresponds to a consistent physical region across images — does not hold under rotation. Third, patch-similarity-based detection is structurally weaker on relational defects (cable_swap) than on local damage or global positional shifts (misplaced, flip), since a relational defect can leave every individual patch looking locally normal.

These findings suggest that architectures aware of long-range positional relationships between components — such as graph-based or transformer-based comparisons of parts to reference layouts — may better address relational defect types, and represent a natural direction for extending this work beyond its current patch-local formulation.

7 Conclusion

This study presented a complete, empirically-validated PatchCore-based anomaly detection pipeline across four MVTec AD categories, achieving image-level AUROC of 0.963, 0.792, 0.935, and 0.947 for metal nut, screw, cable, and transistor respectively. Beyond the numerical results, this work documents two concrete methodological contributions — top- k score aggregation and per-spatial-location calibration — along with a clear empirical account of the conditions under which each is effective. Category-specific limitations (fine scratches, thread defects, relational swaps) were identified, explained mechanistically, and shown to be consistent with the structural assumptions of patch-similarity-based anomaly detection rather than implementation defects. Future work includes extending this study to additional MVTec categories and exploring relational or graph-based scoring mechanisms for the swap/relational

defect class identified here.

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